- *Real-time data transmission and networking*: The introduction of network-centric warfare concepts allowed UAVs to transmit live video, radar imagery, and sensor data to multiple command centers. This enabled real-time intelligence sharing, mission coordination, and target acquisition across military networks.
- *Advanced flight control and redundancy systems*: To ensure operational reliability, third-generation drones employ redundant avionics, triple autopilot systems, and fault-tolerant navigation units, minimizing the risk of mission failure due to single-point system malfunctions.
- *Modular and scalable design*: These UAVs are designed with modular architecture, allowing rapid reconfiguration for specific missions whether for surveillance, strike operations, or environmental monitoring. The modular approach also simplifies maintenance and upgrade cycles.
- *Stealth and low-observable features*: Advanced airframes incorporate radar-absorbing materials, reduced infrared signatures, and optimized aerodynamic profiles, enabling low observability in contested airspaces.
- *Expanded civilian and scientific roles*: Beyond military applications, third-generation drones began to play vital roles in *disaster management*, *border monitoring*, *agricultural mapping*, *and climate research*, marking the diversification of UAV technology into non-combat fields.

4.2 Historical Development of Third-generation Drones

The historical development of third-generation drones marks a transformative period in unmanned aerial vehicle (UAV) evolution, spanning from the early 2000s to the mid-2010s. This era witnessed the